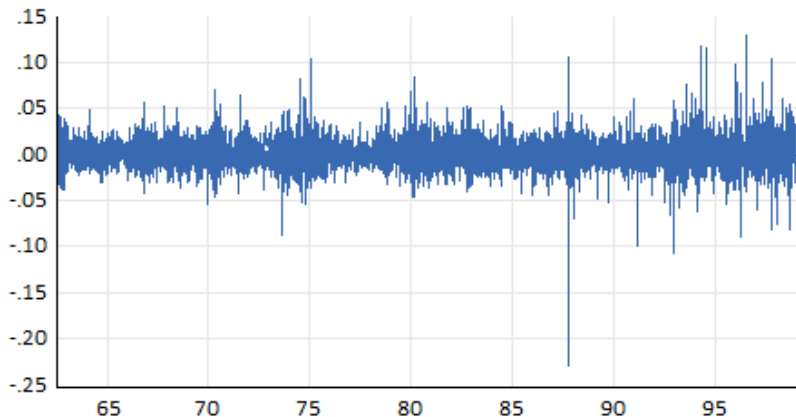


Week 11: IGARCH and ARMA models

Consider the daily log returns of IBM (Y_t) stock from 3 July 1962 to 31 December 1998.



GARCH

Consider the following model:

$$Y_t = c + \varepsilon_t$$

and

$$\varepsilon_t = \sigma_t u_t$$

with a GARCH(1,1) model

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

```
.  
> df <- readxl::read_excel("week 11.xlsx")  
>  
> library(xts)  
> rtn.xts = as.xts(df) # create xts object  
> mean(rtn.xts)  
[1] 0.000556556
```

```

spec_garch11 <- ugarchspec(
  variance.model = list(
    model = "sGARCH",
    garchorder = c(1, 1)
  ),
  mean.model = list(
    armaorder = c(0, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm"
)
fit_garch11 <- ugarchfit(
  spec = spec_garch11,
  data = rtn.xts
)
fit_garch11@fit$matcoef

```

```
> fit_garch11@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	6.942479e-04	1.326736e-04	5.232750	1.670066e-07
omega	2.642554e-06	6.413163e-07	4.120515	3.780265e-05
alpha1	6.180209e-02	5.754673e-03	10.739463	0.000000e+00
beta1	9.288583e-01	6.893984e-03	134.734618	0.000000e+00

IGARCH Models

The estimated GARCH(1,1) model

$$Y_t = 0.0007 + \hat{\varepsilon}_t$$

$$\begin{aligned}\hat{\sigma}_t^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 \hat{\varepsilon}_{t-1}^2 + \hat{\beta}_1 \hat{\sigma}_{t-1}^2 \\ &= 0.0000027 + 0.062 \hat{\varepsilon}_{t-1}^2 + 0.929 \hat{\sigma}_{t-1}^2\end{aligned}$$

Note that the fitted model shows $\hat{\alpha}_1 + \hat{\beta}_1 = 0.062 + 0.929 = 0.991$, which is close to 1.

This phenomenon is commonly observed in practice and it leads to imposing the constraint $\alpha_1 + \beta_1 = 1$ in a GARCH(1,1) model.

Also notice that $\hat{\alpha}_0$ is close to 0.

IGARCH Models

If $\alpha_1 + \beta_1 = 1$ then $\alpha_1 = 1 - \beta_1$.

With $\alpha_1 = 1 - \beta_1$ the GARCH(1,1) model is called the integrated-GARCH (or IGARCH) model:

$$\begin{aligned}\sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \alpha_0 + (1 - \beta_1) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2\end{aligned}$$

Observe that $0 < \beta_1 < 1$.

IGARCH Models

Recall that the unconditional variance of ε_t for an GARCH(1,1) model is

$$\text{var}(\varepsilon_t) = \frac{\alpha_0}{1 - (\alpha_1 + \beta_1)}.$$

For an IGARCH(1,1) model, $\alpha_1 + \beta_1 = 1$, and so $\text{var}(\varepsilon_t)$ does not exist.

This seems hard to justify since it implies that the series is not stationary.

‘Special’ IGARCH Models

It is of particular interest to consider $\alpha_0 = 0$. This is known as an IGARCH(1,1) model.

$$\sigma_t^2 = (1 - \beta_1) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

R will estimate this IGARCH(1,1) model with $\alpha_0 = 0$.

For illustration, we estimate this special IGARCH(1,1) model for the daily log returns of IBM stock

$$Y_t = c + \varepsilon_t$$

and

$$\sigma_t^2 = (1 - \beta_1) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

```

spec_special_igarch <- ugarchspec(
  variance.model = list(
    model = "iGARCH",
    garchorder = c(1, 1)
  ),
  mean.model      = list(
    armaorder = c(0, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm",

  fixed.pars = list(omega = 0)
)

fit_special_igarch <- ugarchfit(
  spec = spec_special_igarch,
  data = rtn.xts
)

fit_special_igarch@fit$matcoef

```

```
> fit_special_igarch@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.0007008544	0.0001343757	5.215635	1.831884e-07
omega	0.0000000000	NA	NA	NA
alpha1	0.0392944376	0.0029877176	13.151992	0.000000e+00
beta1	0.9607055624	NA	NA	NA

IGARCH

The estimated IGARCH(1,1) model is

$$Y_t = 0.000701 + \hat{\varepsilon}_t$$

and

$$\hat{\sigma}_t^2 = (1 - 0.960706) \hat{\varepsilon}_{t-1}^2 + 0.960706 \hat{\sigma}_{t-1}^2$$

J. P. Morgan RiskMetrics

J. P. Morgan developed the RiskMetrics method to VaR calculation.

RiskMetrics assumes that the daily log return (r_t) follows a conditional normal distribution; that is

$$r_t|F_{t-1} \sim N(\mu_t, \sigma_t^2)$$

F_{t-1} is the information set available at time $t - 1$

$\mu_t = E(r_t|F_{t-1})$ is the conditional mean of r_t

σ_t^2 is the conditional variance of r_t

VaR

If we assume that

$$r_t|F_{t-1} \sim N(\mu_t, \sigma_t^2)$$

the VaR_α can be calculated as follows:

Step 1:

Calculate q_α as

$$q_\alpha = \mu_{T+1|T} + \sigma_{T+1} z_\alpha$$

where $\mu_{T+1|T}$ is the 1-step ahead forecast of the conditional mean

$$\mu_{T+1|T} = E(r_{T+1}|F_T)$$

and σ_{T+1}^2 is the 1-step ahead forecast of the conditional variance.

VaR

Step 2:

The $\alpha \times 100\%$ VaR is defined as

$$\text{VaR}_{\alpha} = |W_0 \times q_{\alpha}|$$

RiskMetrics

J. P. Morgan RiskMetrics assumes that the following two quantities evolve over time according to the simple model:

$$\mu_t = 0$$

and

$$\sigma_t^2 = \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) r_{t-1}^2$$

with $0 < \beta_1 < 1$.

Thus RiskMetrics assumes that the conditional mean is zero and σ_t^2 follows an IGARCH(1,1) process.

RiskMetrics

Specifically RiskMetrics assumes that r_t evolves according to the following model

$$r_t = \varepsilon_t$$

and

$$\varepsilon_t = \sigma_t u_t, \quad u_t \sim i.i.d.N(0, 1)$$

$$\sigma_t^2 = \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) r_{t-1}^2$$

When $t = T + 1$, we have

$$\sigma_{T+1}^2 = \beta_1 \sigma_T^2 + (1 - \beta_1) r_T^2.$$

J. P. Morgan RiskMetrics

We use daily log returns of IBM stock (denoted as r_t) to illustrate the calculation of VaR.

Question: What is the 5% VaR over the next trading day on the \$10,000,000 investment in IBM stock using the J. P. Morgan RiskMetrics method?

```

spec_JP_risk <- ugarchspec(
  variance.model = list(
    model = "iGARCH",
    garchorder = c(1, 1)
  ),
  mean.model      = list(
    armaorder = c(0, 0),
    include.mean = FALSE
  ),
  distribution.model = "norm",

  fixed.pars = list(omega = 0)
)

fit_JP_risk <- ugarchfit(
  spec = spec_JP_risk,
  data = rtn.xts
)

fit_JP_risk@fit$matcoef

```

```
> fit_JP_risk@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
omega	0.00000000	NA	NA	NA
alpha1	0.03816874	0.002920518	13.06917	0
beta1	0.96183126	NA	NA	NA

The fitted IGARCH model is

$$\begin{aligned}\hat{\sigma}_t^2 &= \left(1 - \hat{\beta}_1\right) r_{t-1}^2 + \hat{\beta}_1 \hat{\sigma}_{t-1}^2 \\ &= 0.038169 r_{t-1}^2 + 0.961831 \hat{\sigma}_{t-1}^2.\end{aligned}$$

```
# Conditional standard deviation ( $\sigma_t$ )
sigmahat.JP_risk.xts <- sigma(fit_JP_risk)
# Conditional variance ( $\sigma_t^2$ )
sigmahat2.JP_risk.xts <- sigmahat.JP_risk.xts^2
```

```
> tail(sigmahat2.JP_risk.xts)
      [,1]
1998-12-23 0.0004162978
1998-12-24 0.0004090995
1998-12-28 0.0004031099
1998-12-29 0.0003895833
1998-12-30 0.0003795270
1998-12-31 0.0003651936
```

```
> tail(rtn.xts)
      Ret
1998-12-23 0.01509
1998-12-24 0.01588
1998-12-28 0.00698
1998-12-29 -0.01123
1998-12-30 -0.00200
1998-12-31 -0.01272
```

1-step

The sample 1-step ahead forecast is

$$\hat{\sigma}_{T+1}^2 = (1 - \hat{\beta}_1) r_T^2 + \hat{\beta}_1 \hat{\sigma}_T^2$$

At $T = 31/12/1998$, $r_T = -0.01272$ and $\hat{\sigma}_T^2 = 0.000365$. Thus

$$\begin{aligned}\hat{\sigma}_{T+1}^2 &= 0.038169 r_T^2 + 0.961831 \hat{\sigma}_T^2 \\ &= (0.038169) (-0.01272)^2 + 0.961831 (0.000365) \\ &= 0.000357\end{aligned}$$

VaR

What is the 5% VaR over the next trading day on the \$10,000,000 investment based on the J. P. Morgan method?

We calculate $q_{0.05}$ as

$$\begin{aligned} q_{0.05} &= \hat{\mu}_{T+1|T} + \hat{\sigma}_{T+1} z_{\alpha} \\ &= 0 + \sqrt{0.000357} (-1.645) \\ &= -0.031 \end{aligned}$$

Hence the 1-day horizon 5% VaR is

$$\begin{aligned} VaR_{0.05} &= |10,000,000 (-0.031)| \\ &= \$310,000. \end{aligned}$$

AR-GARCH models

The AR(p) models can be used to model the mean equation, and the conditional heteroscedastic models are used to handle the volatility. Suppose r_t follows an AR(p)-GARCH(1,1) model; that is

$$r_t = \mu_t + \varepsilon_t$$

$$\mu_t = c + \phi_1 r_{t-1} + \cdots + \phi_p r_{t-p}$$

with

$$\varepsilon_t = \sigma_t u_t \quad u_t \sim i.i.d.N(0, 1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Example: IBM stock

We now consider the econometric approach to VaR calculation.

Question: What is the 5% VaR over the next trading day on the \$10,000,000 investment in IBM stock using the AR(p)-GARCH(1,1) model?

Example: IBM stock

Consider a restricted AR(2)-GARCH(1,1) model

$$r_t = \phi_0 + \phi_2 r_{t-2} + \varepsilon_t$$

and

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

```

spec_ar2garch11 <- ugarchspec(
  variance.model = list(
    model = "SGARCH",
    garchorder = c(1, 1)
  ),
  mean.model = list(
    armaorder = c(2, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm",
  fixed.pars = list(ar1 = 0)
)

fit_ar2garch11 <- ugarchfit(
  spec = spec_ar2garch11,
  data = rtn.xts
)

fit_ar2garch11@fit$matcoef

```

```
> fit_ar2garch11@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	7.000166e-04	1.296130e-04	5.400820	6.633706e-08
ar1	0.000000e+00	NA	NA	NA
ar2	-2.467268e-02	1.107454e-02	-2.227875	2.588886e-02
omega	2.698798e-06	6.844514e-07	3.943008	8.046585e-05
alpha1	6.242375e-02	6.011161e-03	10.384641	0.000000e+00
beta1	9.279848e-01	7.243114e-03	128.119588	0.000000e+00

Example: IBM stock

The fitted AR(2)-GARCH(1,1) model is

$$r_t = 0.0007 - 0.025r_{t-2} + \hat{\varepsilon}_t$$

and

$$\hat{\sigma}_t^2 = 0.0000027 + 0.0624\hat{\varepsilon}_{t-1}^2 + 0.9280\hat{\sigma}_{t-1}^2$$

```

> res.ar2garch11.xts <- residuals(fit_ar2garch11)

> tail(res.ar2garch11.xts)
      [,1]
1998-12-23  0.015064781
1998-12-24  0.015984559
1998-12-28  0.006635023
1998-12-29 -0.011555486
1998-12-30 -0.002545073
1998-12-31 -0.013714362

> # Conditional variance ( $\sigma_t^2$ )
> sigma2hat.ar2garch11.xts <- sigma(fit_ar2garch11)^2

> tail(sigma2hat.ar2garch11.xts)
      [,1]
1998-12-23 0.0004252821
1998-12-24 0.0004115211
1998-12-28 0.0004005338
1998-12-29 0.0003771362
1998-12-30 0.0003610108
1998-12-31 0.0003381157

```


1-step

The 1-step ahead forecast for the conditional mean $\hat{\mu}_{T+1|T}$ is

$$\begin{aligned}\hat{\mu}_{T+1|T} &= 0.0007 - 0.02467r_{T-1} \\ &= 0.000717 - 0.02467(-0.002) \\ &= 0.00075\end{aligned}$$

The 1-step ahead forecast for the conditional variance $\hat{\sigma}_{T+1}^2$ is

$$\begin{aligned}\hat{\sigma}_{T+1}^2 &= 0.0000027 + 0.0624\hat{\varepsilon}_T^2 + 0.9280\hat{\sigma}_T^2 \\ &= 0.0000027 + 0.0624(-0.0137)^2 + 0.9280(0.000338) \\ &= 0.00033\end{aligned}$$

VaR

What is the 5% VaR over the next day on the \$10,000,000 investment based on the AR(2)-GARCH(1,1) model?

We calculate $q_{0.05}$ as

$$\begin{aligned}q_{\alpha} &= \hat{\mu}_{T+1|T} + \hat{\sigma}_{T+1|T} z_{\alpha} \\&= 0.00075 + \sqrt{0.00033} (-1.645) \\&= -0.0291\end{aligned}$$

Hence the 1-day-ahead 5% VaR is

$$\begin{aligned}VaR_{0.05} &= |10,000,000 (-0.0291)| \\&= \$291,000\end{aligned}$$

We have applied two methods of VaR calculation to the daily log returns of IBM stock on the \$10 million investment.

The results for the 5% VaR over the next trading day are:

1. \$310,000 for the RiskMetrics
2. \$291,000 for a restricted AR(2)-GARCH(1,1) model.

ARMA models

Moving Average Model: MA Models

An alternative way to introduce dynamics into a time series model is by including lags of the disturbance term u_t . A first order moving average model is specified as

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

where u_t is a disturbance term distributed as

$$u_t \sim i.i.d.(0, \sigma_u^2)$$

This model is referred to as MA(1).

MA Models

An MA(2) model is specified as

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2}$$

An MA(q) model is specified as

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2} + \cdots + \theta_q u_{t-q}$$

Consider the MA(1) model

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

Unconditional Mean

The unconditional mean of y_t is found by taking (unconditional) expectations of both sides

$$E(y_t) = \phi_0 + E(u_t) + \theta_1 E(u_{t-1})$$

This expression simplifies by using the property of the disturbance term $E(u_t) = E(u_{t-1}) = 0$, in which case the unconditional mean is

$$\mu = E(y_t) = \phi_0$$

Unconditional Variance

Consider the MA(1) model

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

and its “mean” deviation form

$$\mu = \phi_0$$

Subtracting the second equation from the first enables the model to be rewritten in terms of deviations from the mean as

$$y_t - \mu = u_t + \theta_1 u_{t-1}$$

so the model is reexpressed in deviations from its mean μ .

The unconditional variance is defined as

$$\gamma_0 = E((y_t - \mu)^2)$$

Substituting the expression $y_t - \mu = u_t + \theta_1 u_{t-1}$ into the variance gives

$$\begin{aligned}\gamma_0 &= E((u_t + \theta_1 u_{t-1})^2) \\ &= E(u_t^2 + \theta_1^2 u_{t-1}^2 + 2\theta_1 u_t u_{t-1}) \\ &= E(u_t^2) + \theta_1^2 E(u_{t-1}^2) + 2\theta_1 E(u_t u_{t-1})\end{aligned}$$

As u_t is independent $E(u_t u_{t-1}) = 0$, so

$$\gamma_0 = E(u_t^2) + \theta_1^2 E(u_{t-1}^2)$$

Because u_t is stationary $E(u_t^2) = E(u_{t-1}^2) = \sigma_u^2$, resulting in

$$\gamma_0 = \sigma_u^2 + \theta_1^2 \sigma_u^2 = (1 + \theta_1^2) \sigma_u^2$$

First Order Autocovariance

Defined as

$$\gamma_1 = E((y_t - \mu)(y_{t-1} - \mu))$$

Substituting the expressions $y_t - \mu = u_t + \theta_1 u_{t-1}$ and $y_{t-1} - \mu = u_{t-1} + \theta_1 u_{t-2}$ into γ_1 gives

$$\begin{aligned}\gamma_1 &= E((u_t + \theta_1 u_{t-1})(u_{t-1} + \theta_1 u_{t-2})) \\&= E(u_t u_{t-1} + \theta_1 u_t u_{t-2} + \theta_1 u_{t-1}^2 + \theta_1^2 u_{t-1} u_{t-2}) \\&= E(u_t u_{t-1}) + \theta_1 E(u_t u_{t-2}) + \theta_1 E(u_{t-1}^2) + \theta_1^2 E(u_{t-1} u_{t-2})\end{aligned}$$

First Order Autocovariance

As u_t is white noise $E(u_t u_{t-1}) = E(u_t u_{t-2}) = E(u_{t-1} u_{t-2}) = 0$

$$\gamma_1 = \theta_1 E(u_{t-1}^2)$$

As u_t is stationary $E(u_{t-1}^2) = \sigma_u^2$, then

$$\gamma_1 = \theta_1 \sigma_u^2$$

Second and Higher Order Autocovariance

Defined as

$$\gamma_2 = E((y_t - \mu)(y_{t-2} - \mu))$$

Substituting the expressions $y_t - \mu = u_t + \theta_1 u_{t-1}$ and $y_{t-2} - \mu = u_{t-2} + \theta_1 u_{t-3}$ into γ_2 gives

$$\begin{aligned}\gamma_2 &= E((u_t + \theta_1 u_{t-1})(u_{t-2} + \theta_1 u_{t-3})) \\ &= E(u_t u_{t-2} + \theta_1 u_t u_{t-3} + \theta_1 u_{t-1} u_{t-2} + \theta_1^2 u_{t-1} u_{t-3}) \\ &= E(u_t u_{t-2}) + \theta_1 E(u_t u_{t-3}) + \theta_1 E(u_{t-1} u_{t-2}) + \theta_1^2 E(u_{t-1} u_{t-3}) \\ &= 0\end{aligned}$$

as u_t is independent

$$E(u_t u_{t-2}) = E(u_t u_{t-3}) = E(u_{t-1} u_{t-2}) = E(u_{t-1} u_{t-3}) = 0$$

Hence $\gamma_p = 0, p > 1$.

First Order Autocorrelation

Defined as the ratio of the first order autocovariance of y_t and the variance of y_t .

$$\rho_1 = \frac{\gamma_1}{\gamma_0}$$

For the MA(1) model this is

$$\rho_1 = \frac{\theta_1 \sigma_u^2}{(1 + \theta_1^2) \sigma_u^2} = \frac{\theta_1}{1 + \theta_1^2}$$

Second and Higher Order Autocorrelation

Defined as

$$\rho_2 = \frac{\gamma_2}{\gamma_0} = 0$$

Hence $\rho_p = 0, p > 1$.

Estimation

Estimation of MA models requires a nonlinear algorithm as a result of the moving average component. Assuming that the disturbance term u_t is normally distributed as

$$u_t \sim N(0, \sigma_u^2)$$

the parameters

$$\Theta = \{\phi_0, \theta_1, \theta_2, \dots, \theta_q, \sigma_u^2\}$$

are estimated by maximum likelihood methods. The log-likelihood is

$$\log L = -\frac{1}{2} \log 2\pi - \frac{1}{2} \log \sigma_u^2 - \frac{1}{2\sigma_u^2 T} \sum_{t=1}^T u_t^2$$

where

$$u_t = y_t - \phi_0 - \theta_1 u_{t-1} - \theta_2 u_{t-2} - \dots - \theta_q u_{t-q}$$

Maximize $\log L$ with respect to Θ .

Forecasting

To forecast an MA(1) model at time $T + 1$ write the model as

$$y_{T+1} = \phi_0 + u_{T+1} + \theta_1 u_T$$

Replacing the unknown right-hand side terms by known values gives

$$\hat{y}_{T+1} = \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_T$$

Repeating the steps for $T + 2$ gives the forecast

$$\hat{y}_{T+2} = \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_{T+1} = \hat{\phi}_0$$

as $\hat{u}_{T+1} = 0$.

Forecasting

For $T + H$ the forecast is

$$\hat{y}_{T+H} = \hat{\phi}_0$$

with $H \geq 2$.

Notice that the forecasts from the second period onwards are all the same.

This is a property of the MA(1) model.

The AR(p) model can be extended to allow for moving average dynamics. The ARMA(p,q) model is

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}$$

Consider the ARMA(1,1) model

$$y_t = \phi_0 + \phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}$$

The moments of the ARMA(1,1) model are summarized below

$$\text{Unconditional Mean: } \mu = \frac{\phi_0}{1 - \phi_1}$$

$$\text{Unconditional Variance: } \gamma_0 = (1 + \theta_1^2 + 2\phi_1\theta_1) \frac{\sigma_u^2}{1 - \phi_1^2}$$

$$\text{First order autocovariance: } \gamma_1 = \frac{(1 + \phi_1\theta_1)(\phi_1 + \theta_1)\sigma_u^2}{1 - \phi_1^2}$$

$$\text{First order autocorrelation: } \rho_1 = \frac{(1 + \phi_1\theta_1)(\phi_1 + \theta_1)}{1 + \theta_1^2 + 2\phi_1\theta_1}$$

$$\text{Higher order autocorrelation: } \rho_p = \phi_1 \rho_{p-1}, \quad p \geq 2$$

Application

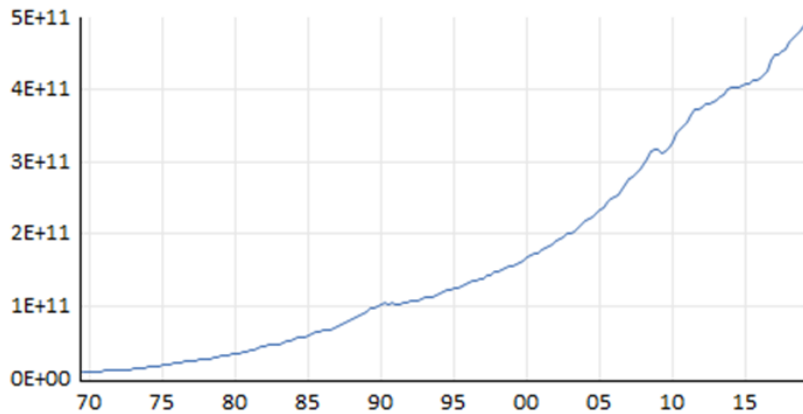
You worked at a Reserve Bank and were tasked with forecasting the GDP growth rate.

The following figure gives time series plots of 2 time series for Australia from 1969Q3 to 2019Q1.

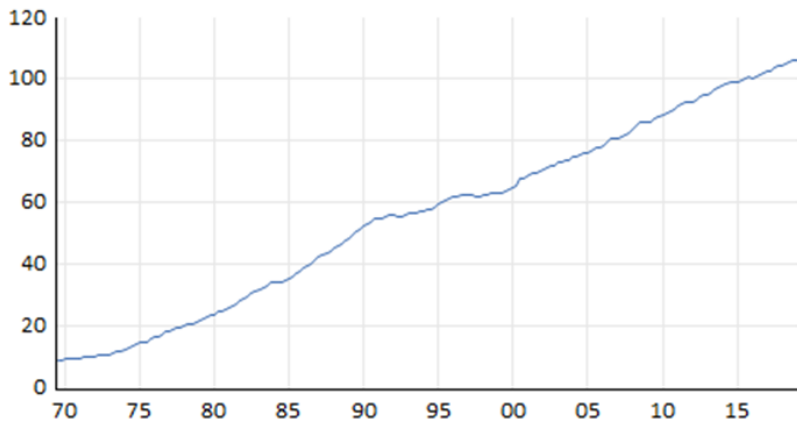
NGDP - Nominal GDP, seasonally adjusted

PRICE - Nominal price (CPI), seasonally adjusted

Nominal GDP



PRICE



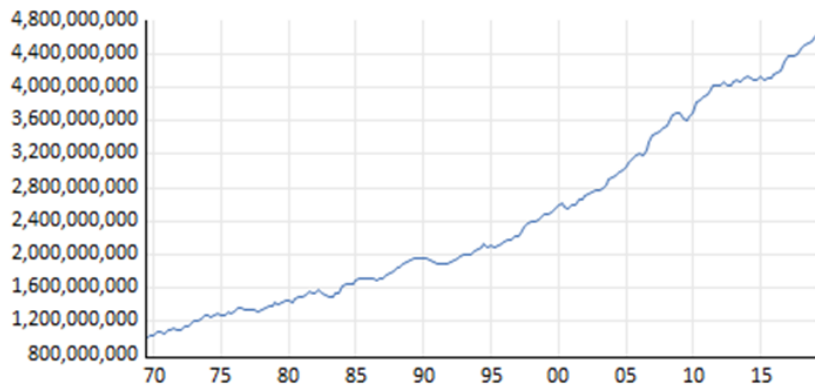
Real GDP

Real GDP is defined as the ratio of nominal GDP (NGDP) and the price level (PRICE)

$$RGDP = \frac{NGDP}{PRICE}$$

Graph shows that RGDP and NGDP both trend upwards with NGDP exhibiting a relatively smoother trend.

Real GDP



Transformation

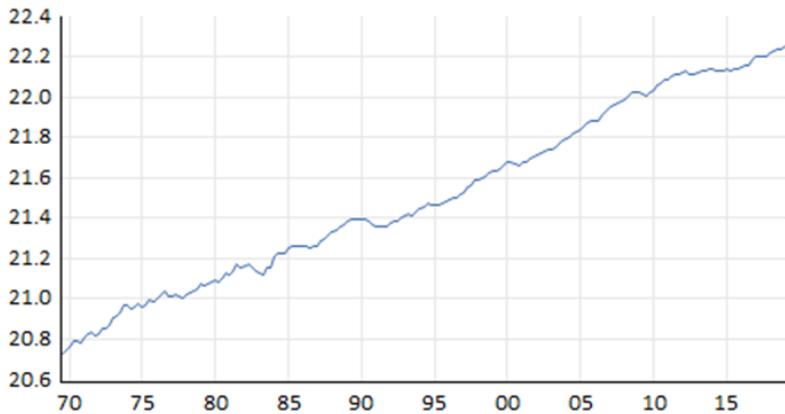
The transformation is $\log(y_t)$.

The graph shows that RGDP has an increasing trend whereas LRGDP has a linear trend.

Log filter has two important characteristics:

- ▶ Capture nonlinearities
- ▶ Can correct for changing volatility (big movements are dampened more than smaller movements)

LN_RGDP



Growth Rate

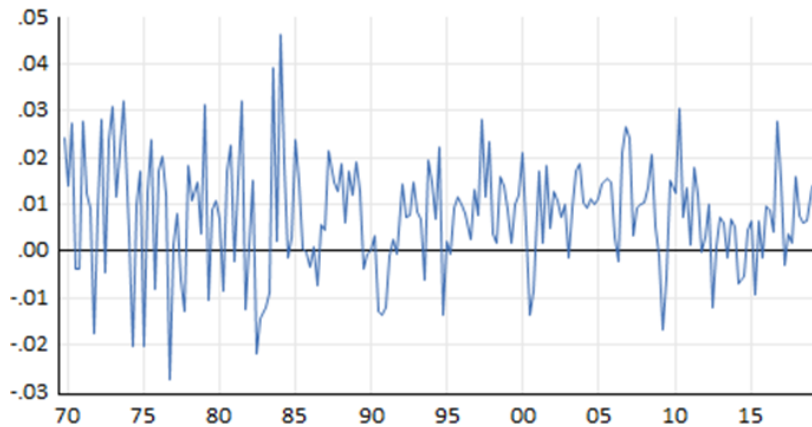
The aim of this application is to specify an ARMA model to forecast the growth rate of real GDP in Australia from 2019Q2 to 2020Q4.

The growth rate is defined as

$$DLRGDP_t = \log(RGDP_t) - \log(RGDP_{t-1})$$

This series is plotted in the next slide.

GROWTH_RATE



```

df <- readxl::read_excel("week 11_rba.xlsx")

library(xts)
n <- nrow(df)
idx_q <- as.Date(as.yearqtr("1969 Q3") + (0:(n - 1)) / 4, frac = 1)
data.xts <- xts(df, order.by = idx_q)
index(data.xts) <- as.yearqtr(index(data.xts))

plot(data.xts$NGDP)
plot(data.xts$PRICE)

rgdp = data.xts$NGDP/data.xts$PRICE

# growth rate
growth_rate <- na.omit(diff(log(rgdp)))

```

ARMA

The results from estimating the ARMA(p,q) for alternative combinations of p and q are given on the next few slides.

```

> # AR(1)
> fit_ar1 <- arima(as.numeric(growth_rate), order = c(1,0,0), include.mean = TRUE)

> fit_ar1

call:
arima(x = as.numeric(growth_rate), order = c(1, 0, 0), include.mean = TRUE)

Coefficients:
          ar1  intercept
      0.1107    0.0077
s.e.  0.0708    0.0010

sigma^2 estimated as 0.000141:  log likelihood = 596.82,  aic = -1187.64

```

```
> fit_ar2 <- arima(as.numeric(growth_rate), order = c(2,0,0), include.mean = TRUE)
```

```
> fit_ar2
```

```
Call:
```

```
arima(x = as.numeric(growth_rate), order = c(2, 0, 0), include.mean = TRUE)
```

```
Coefficients:
```

	ar1	ar2	intercept
	0.1141	-0.0319	0.0077
s.e.	0.0712	0.0711	0.0009

```
sigma^2 estimated as 0.0001409: log likelihood = 596.92, aic = -1185.84
```



```
> fit_arma11 <- arima(as.numeric(growth_rate), order = c(1,0,1), include.mean = TRUE)
> fit_arma11
```

Call:

```
arima(x = as.numeric(growth_rate), order = c(1, 0, 1), include.mean = TRUE)
```

Coefficients:

	ar1	ma1	intercept
	-0.4381	0.5570	0.0077
s.e.	0.3652	0.3364	0.0009

sigma^2 estimated as 0.0001403: log likelihood = 597.31, aic = -1186.63

```
> fit_ma1 <- arima(as.numeric(growth_rate), order = c(0,0,1), include.mean = TRUE)
> fit_ma1
```

Call:

```
arima(x = as.numeric(growth_rate), order = c(0, 0, 1), include.mean = TRUE)
```

Coefficients:

	ma1	intercept
	0.1213	0.0077
s.e.	0.0750	0.0009

sigma^2 estimated as 0.0001409: log likelihood = 596.92, aic = -1187.84

```
> res_ma1 <- residuals(fit_ma1)
> res_ma1.xts <- xts(res_ma1, order.by = index(growth_rate))
> tail(res_ma1.xts)
      [,1]
2017 Q4 -0.005610346
2018 Q1  0.008858538
2018 Q2 -0.001146712
2018 Q3 -0.001418741
2018 Q4 -0.001092497
2019 Q1  0.006526154
```

```
> fit_ma2 <- arima(as.numeric(growth_rate), order = c(0,0,2), include.mean = TRUE)
```

```
> fit_ma2
```

```
Call:
```

```
arima(x = as.numeric(growth_rate), order = c(0, 0, 2), include.mean = TRUE)
```

```
Coefficients:
```

	ma1	ma2	intercept
	0.1235	-0.0405	0.0077
s.e.	0.0714	0.0688	0.0009

```
sigma^2 estimated as 0.0001406: log likelihood = 597.1, aic = -1186.19
```

MA(1)

Let y_t denote the growth rate of real GDP (DLRGDP).

Since lower AIC is better, and -1187.84 is the smallest value, the MA(1) model is preferred by AIC among the models considered.

We consider an MA(1) model to generate forecasts.

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

The fitted MA(1) model

$$y_t = 0.007726 + \hat{u}_t + 0.121308\hat{u}_{t-1}$$

where $\hat{u}_t$ is the residual with variance $\hat{\sigma}_u^2 = 0.000141$.

Forecast

To forecast an MA(1) model at time $T + 1$, we have $\hat{u}_T = 0.0065$ and so

$$\begin{aligned}\hat{y}_{T+1} &= \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_T \\ &= 0.007726 + 0.121308 (0.006526) \\ &= 0.0085177\end{aligned}$$

At time $T + 2$,

$$\begin{aligned}\hat{y}_{T+2} &= \hat{\phi}_0 \\ &= 0.007726\end{aligned}$$

MA(1)

The results of forecasting the MA(1) model from 2019Q2 to 2020Q4 are given below.

Quarter	DLRGDP	Forecast
2019Q1	0.014120	0.014120
2019Q2	NA	0.008518
2019Q3	NA	0.007726
2019Q4	NA	0.007726
2020Q1	NA	0.007726
2020Q2	NA	0.007726
2020Q3	NA	0.007726
2020Q4	NA	0.007726

Forecasts

The forecasts converge to a quarterly growth rate of 0.007726. This represents an annualized growth rate of

$$4 \times 0.007726 = 0.030904$$

or 3.09% per annum.

The following table contains the IMF projections of real GDP (annual percentage growth rates) for 2021 and 2022.

Latest World Economic Outlook Growth Projections

(real GDP, annual percent change)	PROJECTIONS		
	2020	2021	2022
World Output	-3.1	5.9	4.9
Advanced Economies	-4.5	5.2	4.5
United States	-3.4	6.0	5.2
Euro Area	-6.3	5.0	4.3
Germany	-4.6	3.1	4.6
France	-8.0	6.3	3.9
Italy	-8.9	5.8	4.2
Spain	-10.8	5.7	6.4
Japan	-4.6	2.4	3.2
United Kingdom	-9.8	6.8	5.0
Canada	-5.3	5.7	4.9
Other Advanced Economies	-1.9	4.6	3.7

Source: World Economic Outlook, October 2021

Term Spread

U.S. GDP

```
library(xts)

df <- readxl::read_excel("week 11 Term spread.xlsx")

# Rename first column
names(df)[1] <- "Quarter"

# Convert quarterly labels such as 1960:01 into year-quarter format
year <- as.numeric(substr(df$Quarter, 1, 4))
quarter <- as.numeric(substr(df$Quarter, 6, 7))

qtr_index <- as.yearqtr(year + (quarter - 1) / 4)

# Create xts object with quarterly index
macro.xts <- xts(
  as.data.frame(df[, c("GS10", "TB3MS", "RSPREAD", "RECESSION", "GDP")]),
  order.by = qtr_index
)
```

```
> tail(macro.xts)
```

		GS10	TB3MS	RSPREAD	RECESSION	GDP
2016	Q3	1.563333	0.2966667	1.266667	0	16778.15
2016	Q4	2.130000	0.4300000	1.700000	0	16851.42
2017	Q1	2.443333	0.5900000	1.853333	0	16903.24
2017	Q2	2.263333	0.8900000	1.373333	0	17031.08
2017	Q3	2.243333	1.0366667	1.206667	0	17163.89
2017	Q4	2.370000	1.2066667	1.163333	0	17271.70

GS10 is 10-Year Interest Rate

TB3MS is 3-Month Interest Rate

Rspread is GS10-TB3MS

Recession is an indicator

GDP is U.S. GDP (billions of \$2009)

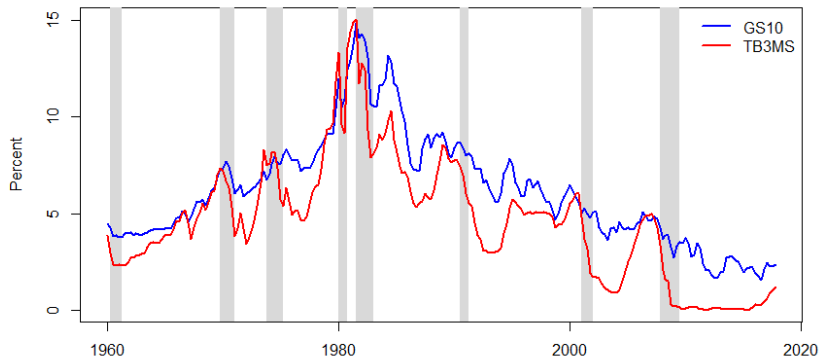
Forecasting GDP Growth Using the Term Spread

Interest rates on long-term and short-term bonds move together but not one for one.

The figure plots interest rates on 10-year U.S. Treasury bonds and 3-month Treasury bills from 1960 through 2017.

These interest rates show the same long-run tendencies: Both were low in the 1960s, both rose through the 1970s and peaked in the early 1980s, and both fell subsequently.

GS10 and TB3MS



Term Spread

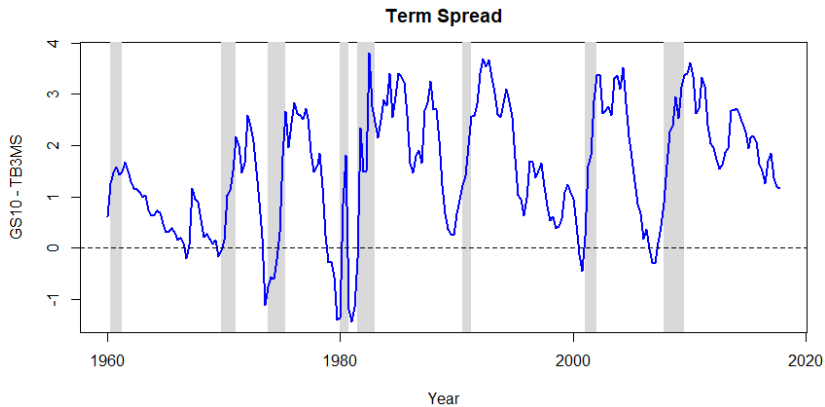
But the gap, or difference, between the two interest rates has not been constant:

While short-term rates are generally below long-term rates, the gap between them narrows and even disappears shortly before the start of a recession.

Recessions are shown as the shaded bars in the figure.

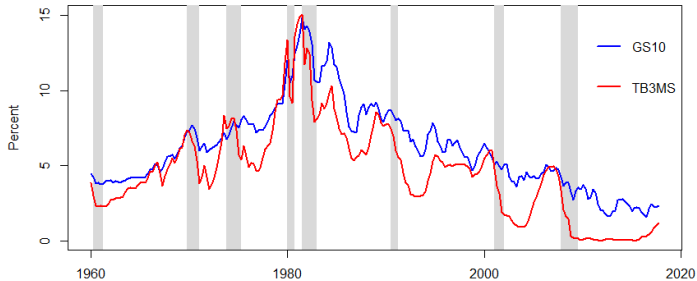
This difference between long-term and short-term interest rates is called the term spread and is plotted in the next slide.

The term spread is generally positive, but it falls toward or below 0 before recessions.

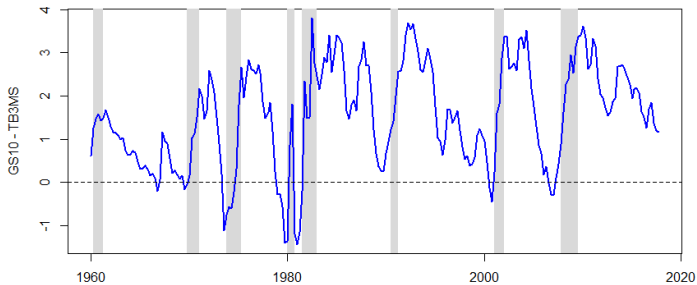


The Figure suggests that the term spread might contain information about the future GDP growth that is not already contained in past values of GDP growth.

GS10 and TB3MS



Term Spread



Construct GDP growth rate.

```
> macro.xts$GDPGR <- diff(log(macro.xts$GDP)) * 400  
> tail(macro.xts)
```

		GS10	TB3MS	RSPREAD	RECESSION	GDP	GDPGR
2016	Q3	1.563333	0.2966667	1.266667	0	16778.15	2.742267
2016	Q4	2.130000	0.4300000	1.700000	0	16851.42	1.743040
2017	Q1	2.443333	0.5900000	1.853333	0	16903.24	1.228157
2017	Q2	2.263333	0.8900000	1.373333	0	17031.08	3.013954
2017	Q3	2.243333	1.0366667	1.206667	0	17163.89	3.107115
2017	Q4	2.370000	1.2066667	1.163333	0	17271.70	2.504579

We use data starting in 1962:Q1 and reserve the final observation, 2017:Q4, to illustrate computing the forecast and forecast error. Thus the estimation period is from 1962:Q1–2017:Q3.

The conjecture is readily checked by augmenting the AR(2) model to include the first lag of the term spread:

$$\widehat{GDPGR}_t = 0.94 + 0.27 GDPGR_{t-1} + 0.19 GDPGR_{t-2} + 0.42 TSpread_{t-1}.$$

The forecast of the rate of GDP growth in 2017:Q4 is obtained by substituting the 2017:Q2 (which is 3.01) and 2017:Q3 (which is 3.11) values of GDP growth, along with the value of the term spread in 2017:Q3 (which is 1.21); the resulting forecast is

$$\widehat{GDPGR}_{2017:Q4|2017:Q3} = 2.9\%,$$

and the forecast error is -0.4% .

```
# Create variables
GDPGR <- macro.xts$GDPGR
TSspread <- macro.xts$RSPREAD

# Create lagged variables
reg_data <- merge(
  GDPGR,
  lag(GDPGR, 1),
  lag(GDPGR, 2),
  lag(TSspread, 1)
)

# Rename columns explicitly
colnames(reg_data) <- c( "GDPGR", "GDPGR_lag1",
  "GDPGR_lag2",
  "TSspread_lag1"
)

# Remove missing values caused by differencing and lags
reg_data <- na.omit(reg_data)
```

```
# Subset from 1962 Q1 to 2017 Q3
reg_data <- reg_data[
  index(reg_data) >= as.yearqtr("1962 Q1", format = "%Y Q%q") &
  index(reg_data) <= as.yearqtr("2017 Q3", format = "%Y Q%q")
]

# Estimate the model
fit <- lm(
  GDPGR ~ GDPGR_lag1 + GDPGR_lag2 + Tspread_lag1,
  data = as.data.frame(reg_data)
)

summary(fit)
```

```
> summary(fit)
```

Call:
lm(formula = GDPGR ~ GDPGR_lag1 + GDPGR_lag2 + Tspread_lag1,
 data = as.data.frame(reg_data))

Residuals:

Min	1Q	Median	3Q	Max
-10.3428	-1.6306	-0.1521	1.6320	13.2549

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.94082	0.39733	2.368	0.01876	*
GDPGR_lag1	0.26806	0.06546	4.095	5.95e-05	***
GDPGR_lag2	0.18975	0.06534	2.904	0.00406	**
Tspread_lag1	0.42142	0.16626	2.535	0.01195	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.976 on 219 degrees of freedom
Multiple R-squared: 0.1697, Adjusted R-squared: 0.1583
F-statistic: 14.92 on 3 and 219 DF, p-value: 7.14e-09

Calculation

The forecast error is defined as

$$\text{Forecast error} = GDPGR_{2017:Q4} - \widehat{GDPGR}_{2017:Q4|2017:Q3}.$$

Using the given values,

$$\begin{aligned}\widehat{GDPGR}_{2017:Q4|2017:Q3} &= 0.94 + 0.27(3.11) + 0.19(3.01) + 0.42(1.21) \\ &\approx 2.9\%.\end{aligned}$$

The actual GDP growth in 2017:Q4 is 2.5%, then the forecast error is

$$\text{Forecast error} = 2.5 - 2.8598 = -0.3598 \approx -0.4.$$

We say the forecast error is approximately

−0.4 percentage points.